SHARDA UNIVERSITY



Student Project Diary
Session : 2026-27
Course Name : Capstone-1
Course Code : CSP497

Title of The Project :
Intelligent DevOps Monitoring & Incident Response System

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Name of the Guide: Ms. Aliza Rana (Assistant Professor)

Project Diary (Student)

Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
One	22 Jul 2026	First meeting; project group formed		Finalise topic; study existing monitoring tools (Prometheus, Nagios, Datadog)	
Two	29 Jul 2026	Topic finalised; existing tools studied		Literature survey: AIOps, anomaly detection, LLM incident analysis	
Three	05 Aug 2026	Literature survey done; gaps noted		Write problem statement, objectives and scope	
Four	12 Aug 2026	Problem statement and objectives done		Requirement analysis; select tools (Docker, Prometheus, Ollama)	
Five	19 Aug 2026	Requirements and tool list ready		Design system architecture: monitor, detect, analyze, respond, verify	
Six	26 Aug 2026	Architecture and plan drafted		Design IncidentEvent / ResponsePlan schemas, alert rules, safety policies	
Seven	02 Sep 2026	Schemas, rules and policies designed		Set up Docker and Prometheus; base Flask app with / metrics	

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Eight	09 Sep 2026	Environment set up; base app monitored by Prometheus		Add fault injection: CPU, memory, latency, error faults	
Nine	16 Sep 2026	Fault injection working		Draft Invention Disclosure Form (patent)	
Ten	23 Sep 2026	IDF first draft written		System redesign: old design used only fixed runbooks; add LLM root-cause analysis, safety checks, re-analysis loop	
Eleven	30 Sep 2026	Redesign done: fixed-runbook design replaced by LLM analysis with safety checks and verification		Multi-service app (frontend, API, Redis), alert rules, fault injector	
Twelve	07 Oct 2026	Done; also built event gateway, LLM analyzer, responder and verifier		Web dashboard (Next.js), ML anomaly detector; finalise IDF	
Thirteen	14 Oct 2026	Dashboard, ML detector and final IDF done; 9 scenarios live		Evaluation runner; stage 1 evaluation of 9 fault scenarios	
Fourteen					
Fifteen					
Sixteen					

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Seventeen					
Eighteen					
Nineteen					
Twenty					

Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: One **Date of Visit:** 22 Jul 2026

Brief Description of task to be completed till next Visit:

Finalise topic; study existing monitoring tools (Prometheus, Nagios, Datadog)

Details of approaching the task and solving the task:

Listed possible topics and chose intelligent DevOps monitoring, because handling an alert is still mostly manual work. Studied Prometheus (metrics and alert rules), Nagios (host and service checks) and Datadog (hosted monitoring). Noted that all three raise alerts but leave the diagnosis and the fix to an engineer.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Two **Date of Visit:** 29 Jul 2026

Brief Description of task to be completed till next Visit:

Literature survey: AIOps, anomaly detection, LLM incident analysis

Details of approaching the task and solving the task:

Read papers and articles on AIOps, anomaly detection (Isolation Forest, z-scores), root-cause analysis in microservices and LLM-based incident analysis. Recorded the gaps: most tools only detect problems, automated fixes have no safety check, and nothing confirms that a fix actually worked.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Three **Date of Visit:** 05 Aug 2026

Brief Description of task to be completed till next Visit:

Write problem statement, objectives and scope

Details of approaching the task and solving the task:

Wrote the problem statement: the time between an alert and its resolution is long because the analysis is manual. Set the objectives: detect faults, find the root-cause service, fix it safely and verify the fix. Fixed the scope to Docker Compose on a laptop with metrics only; Kubernetes is out of scope.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Four **Date of Visit:** 12 Aug 2026

Brief Description of task to be completed till next Visit:

Requirement analysis; select tools (Docker, Prometheus, Ollama)

Details of approaching the task and solving the task:

Listed the functional requirements (collect metrics, detect faults, analyze, respond, verify, show a dashboard) and the non-functional ones (runs on a laptop, safe, explainable). Selected Docker Compose, Prometheus with Alertmanager and cAdvisor, Python (Flask, FastAPI) and Ollama with a local Llama 3.1 8B model.



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Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Five **Date of Visit:** 19 Aug 2026

Brief Description of task to be completed till next Visit:

Design system architecture: monitor, detect, analyze, respond, verify

Details of approaching the task and solving the task:

Drew the pipeline Monitor, Detect, Analyze, Decide, Respond and Verify. Placed an event gateway between the alerts and the analysis, and made the responder the only component allowed to change the system. Compared an ML-centric plan with an LLM-centric plan before choosing.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Six **Date of Visit:** 26 Aug 2026

Brief Description of task to be completed till next Visit:

Design IncidentEvent / ResponsePlan schemas, alert rules, safety policies

Details of approaching the task and solving the task:

Defined the IncidentEvent schema (alerts plus metric context) and the ResponsePlan schema (root cause, actions, confidence). Drafted alert rules for CPU, memory, latency, errors and service down. Wrote the safety policies: allowlisted actions, known targets, confidence of at least 0.7, a 5-minute cooldown, at most 3 actions per incident and approval for risky actions.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Seven **Date of Visit:** 02 Sep 2026

Brief Description of task to be completed till next Visit:

Set up Docker and Prometheus; base Flask app with /metrics

Details of approaching the task and solving the task:

Installed Docker Desktop and wrote a Compose file with Prometheus. Built a Flask test app that exposes /metrics with request counts and latency histograms, and checked in the Prometheus targets page that it was being scraped.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

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Week No: Eight **Date of Visit:** 09 Sep 2026

Brief Description of task to be completed till next Visit:

Add fault injection: CPU, memory, latency, error faults

Details of approaching the task and solving the task:

Added /inject and /reset endpoints for CPU, memory, latency and error faults; each fault clears itself after its duration. Made the memory fault write to its pages so that container memory really rises. Exported fault_*_active gauges as ground truth, used only for scoring.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Nine **Date of Visit:** 16 Sep 2026

Brief Description of task to be completed till next Visit:

Draft Invention Disclosure Form (patent)

Details of approaching the task and solving the task:

Wrote the Invention Disclosure Form: the problem, the prior art (monitoring systems, rule-based alerting, manual incident response), the proposed system and its novel parts. Prepared the block diagrams.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Ten **Date of Visit:** 23 Sep 2026

Brief Description of task to be completed till next Visit:

System redesign: old design used only fixed runbooks; add LLM root-cause analysis, safety checks, re-analysis loop

Details of approaching the task and solving the task:

Reviewed the first design: it relied on fixed runbooks and its LLM only wrote summaries, so unseen faults could not be handled and a failed fix was never re-analyzed. Redesigned it so the LLM finds the root cause and proposes actions, a policy validator checks every action before it runs, and a verifier re-checks and re-analyzes up to 3 times before escalating. Kept the runbooks as a fallback and moved from Kubernetes to Docker Compose.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Eleven **Date of Visit:** 30 Sep 2026

Brief Description of task to be completed till next Visit:

Multi-service app (frontend, API, Redis), alert rules, fault injector

Details of approaching the task and solving the task:

Split the app into frontend, api-service and Redis, with redis_exporter and cAdvisor. Wrote Prometheus alert rules with 30 s windows (about 40 s to detect) and an Alertmanager webhook. Built the fault injector with 9 scenarios that logs ground truth, plus Locust load. Also built the event gateway, the LLM analyzer with runbook fallback, and the responder with verifier, and ran the full loop on real containers.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Twelve **Date of Visit:** 07 Oct 2026

Brief Description of task to be completed till next Visit:

Web dashboard (Next.js), ML anomaly detector; finalise IDF

Details of approaching the task and solving the task:

Built a Next.js dashboard with a service map, incident timelines, approvals and statistics. Recorded 78 minutes of normal traffic and trained the ML detector (robust z-scores plus Isolation Forest) to catch faults no rule covers, such as a slow memory leak. Finalised the IDF with claims and figures. Ran all 9 scenarios live: 9 of 9 detected.



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Name: Muhammad Shayaan Ali **Enrollment No. :** 2301010569

Project Title: Intelligent DevOps Monitoring & Incident Response System

Week No: Thirteen **Date of Visit:** 14 Oct 2026

Brief Description of task to be completed till next Visit:

Evaluation runner; stage 1 evaluation of 9 fault scenarios

Details of approaching the task and solving the task:

Wrote an evaluation runner that injects each scenario, switches between the runbook-only baseline, the LLM baseline and the full system, and scores the results. Stage 1 (81 runs): 100% of faults detected and 0 unsafe actions; average detection time 27 s for the full system against 52 s for the runbook-only baseline.



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